

Rational Altruist

Adventures of a would-be do-gooder.

Replaceability

by paulfchristiano

Suppose I am trying to evaluate the consequences of taking job X. Here is a sequence of (hopefully decreasingly) naive ways to think about the impact of my decision.

(I don't know if the later analyses are actually reasonable, but I'd really like to see more intellectually serious discussion of this issue by people who understand the world, and particularly economics, better than I do. I wouldn't be surprised if more sophisticated versions of this analysis are already well-understood amongst economists, and simply haven't been noticed by altruists trying to understand this issue. In that case, hopefully someone can point that out to me. Some of these analyses, and more sophisticated elaborations on them, appear in Ben Todd's masters thesis (http://www.academia.edu/1807196/Which_Ethical_Careers_Make_a_Difference_The_Replaceability_Issue_in_the_Ethics_of_Career_Choice).)

(Skip to section 6 if you want the punchline.)

1.

I should look at the direct consequences of doing job X, and choose a job for which those consequences are best.

2.

If I don't take job X someone else will. So I shouldn't be at all concerned with the immediate consequences of a job. Instead I should focus on the money I earn doing job X, which I will be able to spend at my own discretion (and which a substitute would have spent differently).

2b.

I should also reason about other parts of the job which I might do differently from my substitute; for example, if I intend to sabotage a company, my substitute probably won't. If I will do a better or worse job than my substitute, that will also be relevant, etc.

3.

But when I take job X I slightly increase the supply of people who are willing to do job X, slightly decrease the wages for job X, and therefore slightly increase the amount of that job X that gets done. So I should compare the elasticity of supply and demand in this market to determine what will happen to the total amount of job X that gets done. (I think this is a standard calculation in basic microeconomics.)

3b.

Markets aren't perfect spheres, and wages, demands, supplies, etc., will be slow and inconsistent in their reaction to an increase in supply. But if we look at the aggregate of many such decisions over a long period of time, this analysis should hold on average over many such decisions (and that should be good enough for our decision). Similarly, because workers commit to industries (for example by obtaining a degree in neuroscience) and industries commit to a number of workers (for example by investment in complementary capital) there is some necessary lag between an increased wage for job X and an increased supply. But over a long time period, on average, these effects seem to be neutral.

4.

In most industries the supply of labor is very elastic (because at equilibrium, if wages within an industry are lowered, people will instead take higher paying jobs elsewhere). Thus if I take job X, it will have little effect on the total amount of job X that gets done. So I should reason entirely about the money.

5.

But when I displace someone who would have done job X they take job Y, thus displacing someone else

who goes on to take job Z. This process continues, each step having a small impact on wages, but the aggregate impact of many steps being unpredictable. To analyze the outcome we should appeal to a simple equilibrium analysis: there is some supply of labor, and both before and after my introduction to the workforce, that supply of labor was being used roughly efficiently. So my impact is the same regardless of what job X is—it is *as if* I were doing the most economically productive activity (on the margin).

5b.

Incentives aren't defined in terms of economic productivity. For example, there may be philanthropists or governments funding activity in a domain. Then we need to adjust the analysis to say: my impact is *as if* I were doing the activity which is most rewarded on the margin.

5c.

Different employees aren't perfect substitutes. I have my own set of skills, and the above analysis actually shows that I would be doing the most effective thing for a worker with my skills. So if I am a researcher, my impact will be to do the most productive research.

But of course it doesn't matter what my skills are, it matters what skills I am using—the above analysis needs to be modified to take into account the fact that I might not pursue the most lucrative opportunity for someone with my skills. If I have a few different skills which don't often appear together, I can choose which of those skills to use (and therefore change my impact).

However, if there are any laborers who were previously ambivalent between using those skills, then this effect will disappear. And if I can do either job X or job Y, in a large world it is very likely that others can as well. So I can actually have an impact only when everyone else faced with the choice between job X and job Y would definitely choose one or the other, i.e. when one of the jobs is less desirable than the other.

6.

But laborers differ in more ways than just their skills. The last consideration was actually a special case of the more general observation that laborers differ in terms of the *values* they have. In fact I am not alone in considering jobs based on more than just my anticipated compensation—if I would choose job X over job Y because of altruistic considerations, there may be others who would do the same.

What I am really doing when I decide to choose my job on the basis of altruistic concerns in addition to monetary compensation is changing the *aggregate demand* of laborers. In a world where all laborers made their decisions on the basis of only monetary compensation, the jobs that got done would be those that produced the most economic value. In a world where all laborers made their decisions on the basis of altruistic impact, the jobs that got done would be those that produced the most (direct) altruistic impact.

In intermediate worlds, in which the labor force collectively makes its decisions on the basis of a combination of altruistic impact and economic value, the jobs that get done are those that are optimal according to that combination of values.

Assuming that workers all value money enough that markets continue to function efficiently and facilitate (implicit) trade amongst people with different values, then my decision to base my job selection on a particular combination of money and other values serves to (slightly) change the collective values of the labor force, which in turn slightly changes which jobs get done.

6b.

If my values are sufficiently unusual that I am taking jobs which no one else would consider, then the above analysis breaks down—unless individuals can engage in explicit trade with one another, in which case it seems to go through (modulo uncertainty about bargaining outcomes). This seems to be the role of money in the above picture.

Appendix:

An analogous situation seems to obtain with respect to spending money, charitable or otherwise (and in fact in a broader set of domains). If I choose to spend my money to accomplish project X, others may also stop spending money on project X. The proper way to understand my spending seems to be as a change to the *aggregate demands* of people with money. If I am spending money on things that are good for me, then I've shifted the notion of "economic value" to include things that are good for me. If I spend more money, I shift it farther. If I spend money on things that are good for future generations, then I've shifted the notion of "economic value" to include things that are good for future generations.

So if we are comparing job-related impacts of a career with donation-related impacts, it seems like we should be reasoning about the importance of shifting "what people are willing to work for" vs. "what people are willing to pay for." I hope that this kind of framework can lead to a less brittle picture than we currently have.

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14 Comments to "Replaceability"

Taxonomy of change « Rational Altruist says:

February 8, 2013 at 2:57 am

[...] into the world of tomorrow—I'd rather just wait a year. Unfortunately, in light of concerns about replaceability, many of our actions may (essentially) have the effect of accelerating progress in one

domain or [...]

Reply

Robert Jack Will says:

February 17, 2013 at 12:58 pm

Hi Paul,

it's an interesting analysis that you are making here and as I understand it the difference between choosing one job over another (or donating to one charity instead of another) is always smaller than it would naively seem, because the larger system around us fills gaps that would appear if we were not there. (I think that for many rationalists the only hope is that many other people will decide using a similar algorithm, therefore making small differences accumulate.)

In a way it seems that any job we'll take, we're only a little cog in a larger machine and that's not a thought that really motivates people to act and define their careers.

So what else can we use as a guiding post? Two different things come to my mind. First, instead of changing the world by what we do (that is, our jobs), we can change the world by how we do it (that is, our values). If someone brilliantly embodies the values of rationality and empathy, they will radiate their influence to all the people who are close to them in work or private life. Some of those people might be much more influential than you are. For example, even if I have only little influence over my direct manager, she values my opinion much and sometimes it influences her actions thus spreading my values out into the entire team or beyond. Or in private life, one of your friends and/or neighbors could be a very influential engineer, doctor, researcher who'll occasionally discuss with you some of the tough question they're facing in their highly influential job. If you have the empathy to win this person's trust and the rationality to offer him progressive insight, you're exerting important influence on society.

From this point of view, the best way to choose a job is to look at things which you like doing (concentrating on the "flow" of doing instead of just the outcome) and which make good use of your skills and helps you improve your skills. Get pleasure and flow from your job, while getting meaning from spreading your values and indirect influence.

The other thing is simply that the most influential jobs in society are leadership jobs. If people without ambition take on any kind of leadership position, they will just be a kind of very large cog in the machine of society, but still a cog. (I think of G.W.Bush here, who was the puppet of his more ambitious and more influential allies.)

However, the key difference is that if you do have a clear ambition (like spreading your values, solving a specific problem), then the difference between you and somebody else getting the leadership position will be huge.

From what I see on LessWrong.com most rationalists aren't eager to climb up the career ladder to leadership positions because they are perceived as just being larger cogs in the machine and require a lot of "conformance".

In any case I think that in order to create consistent change, one needs to train and develop their empathy to be able to reach out to all kinds of people, especially the non-rationalist, conformant masses.

Personally, I studied Computer Science as a kid "just because I liked it" and then later as a more strategically thinking adult, I chose to start my career in IT because it would develop my skills best, make me happy and balanced and thus allow me to radiate positive influence.

Recently I got back in touch with rationalism and "strategic altruism" and I gained much insight from reading the book "search inside yourself" and practicing its advice every day. While the IT job gives me

a solid happy base of living, my training in empathy (and scheming, as taught in other books) allows me to increase my influence. I might switch industries later, to use my influence for the advance of a specific cause, but right now, I am still training my empathy and leadership skills.

(One of the influential scheming books is “Switch” by the Heath brothers.)

REPLY

Peli Grietzer says:

February 22, 2013 at 7:31 pm

Point 6 deals with the a-causal rather than the causal impact of your choice, right? Or am I misreading?

REPLY

paulfchristiano says:

February 22, 2013 at 11:54 pm

No, point 6 is intended to deal with the causal impact. I am talking about causal impacts using this (unusual) language because the system’s dynamics are too complicated, and so I need to fall back to equilibrium analysis. My choice of career has a causal impact which is modulated by substitution effects, and I am trying to indirectly characterize that causal impact.

REPLY

Economics of small changes | Rational Altruist says:

May 2, 2013 at 10:39 pm

[...] The above analysis suggests you should care about the long run elasticity, even if the change is temporary (at least if you mostly care about the long run). This is just because the effects of changes in supply are probably linear, and a temporary change is just a small fraction of long-term change. I implicitly appealed to this principle in my earlier post about replaceability. [...]

Reply

Making people richer | Rational Altruist says:

May 7, 2013 at 12:07 am

[...] think there is a broad class of goods—namely, stuff people would buy anyway—for which replaceability significantly “washes out” the difference between different ways people could become [...]

Reply

Beware brittle arguments | Rational Altruist says:

May 8, 2013 at 8:30 pm

[...] same resources and people will often push the same kinds of developments (so replacement mixes the effects faster, and your impact ends up being the same as the average development of [...]

Reply

Contributing to tech progress | Rational Altruist says:

May 26, 2013 at 6:05 pm

[...] course in reality this isn’t literally what is going on. The person who takes a job on project Q is very likely to pass up a quite similar job. But someone [...]

Reply

Altruism and profit | Rational Altruist says:

July 11, 2013 at 11:28 am

[...] Replaceability seems to be a serious consideration in altruistic activities. When working on a

project that would be done anyway, it is tempting to reason “this will be done anyway in one year, so I should be thinking about the impact of doing it one year sooner.” This may be an improvement over the naive view, but it's still not the complete picture. If you complete a high-impact technology project which other technologists would have tackled in a year, the counterfactual doesn't just involve that project finishing a year sooner—it also involves those technologists going off to do whatever else they would have done. [...]

Reply

Show Me the Harm | The Centre for Effective Altruism

says:

September 3, 2013 at 5:28 pm

[...] than the average banker. By (ii), I mean that if someone turns down a job at Goldman, Goldman will take some actions to compensate for that loss, for instance they'll hire other people. This will probably result in a significant proportion of [...]

Reply

Show me the harm | 80,000 Hours says:

September 1, 2014 at 1:00 pm

[...] than the average banker.⁵ By (ii), I mean that if someone turns down a job at Goldman, Goldman will take some actions to compensate for that loss, for instance they'll hire other people. This will probably result in a significant [...]

Reply

What does economics tell us about replaceability? | 80,000 Hours

says:

September 1, 2014 at 1:24 pm

[...] reasoning in general (read these first if you're not familiar with replaceability!); Paul Christiano and Ben Kuhn (among others) have written blogs further exploring the concept, and its various [...]

Reply

My Initial Post-Grad Plans | Impactful Work says:

June 29, 2015 at 6:29 pm

[...] See this post on replaceability in the context of job choice. [...]

Reply

'Replaceability' isn't as important as you might think (or we've suggested) - 80,000 Hours

says:

July 27, 2015 at 2:16 pm

[...] 2012, the year 80,000 Hours was founded. And as we learned more, we became more cautious (e.g. see this post and this one). Due to this uncertainty, replaceability never played a big part in our online career [...]

Reply